

SHALIKA KUMBHAM

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EDUCATION

University of North Carolina at Chapel Hill

Aug 2024 - Dec 2025

M.S, Computer Science

4.0/4.0

Indian Institute of Technology, Kharagpur

July 2018 - April 2022

B.Tech, Mechanical Engineering

8.45/10

Coursework: Data Structures and Algorithm, Computer Architecture and Operating Systems, Machine Learning(ML), Natural Language Processing(NLP), Complex Networks

PUBLICATIONS

Shalika Kumbham, Anvesh Rao Vijjini, Snigdha Chaturvedi - “*Guided Relational Analysis Improves Social Intelligence in LLMs*” - ACL2026 (Under review)

Shalika Kumbham, Abhijit Debnath, Krothapalli Sreenivasa Rao. “*Efficient Indexing of Meta-Data (extracted from educational videos)*” - [PrePrint](#)

SKILLS

Programming Languages: C++, C, Python, Java, JavaScript, SQL

OracleDB, Linux, PostgreSQL, React, Node.js, Django, GitHub, CI/CD, Agile, FastAPI, Flask, Redis, Kafka, Pytorch

INDUSTRIAL EXPERIENCE

Software Development Engineer | Oracle India Pvt. Ltd.

July 2022 – Aug 2024

- Engineered high-performance worksheet infra, optimizing 500+ calculations **across 300 columns under 30ms**.
- Reduced REST API **response times by 50%** through parallelization, scaling infra for high-traffic use.
- Developed Peer-appreciation app (database design, PL/SQL, architecture) supporting **millions of transactions**.

Machine Learning Research Intern(Computer Vision) | FelixSolutions AI

June 2021 – Aug 2021

- Built end-to-end data pipeline** for forgery detection; including data ingestion and model readiness.
- Experimented with CNN and LSTM models for multimodal forgery detection, **achieving 75% accuracy**.
- Automated evaluation** with F1 score, improving model robustness and reliability.

KEY PROJECTS

Apple AirTag Assistant(RAG-based Chatbot)

Dec 2025 - March 2026

- Built a **LangChain-based RAG** chatbot with chunking, embeddings, and hybrid (vector + lexical) retrieval.
- Designed scalable retrieval using Supabase (pgvector) and Cloudflare Workers AI for **low-latency** LLM inference.
- Developed and deployed a **Next.js (TypeScript)** app with citation-backed answers and **modular architecture**.

TryOnDiffusion Integration Project

Aug 2025 - Dec 2025

- Implemented **diffusion-based models** to generate photorealistic try-on previews for fashion e-commerce.
- Containerized the pipeline using **Docker**, orchestrated with **Kubernetes**, and deployed on **AWS**.

In-Context Learning in Snake and Ladders Game

Jan 2025 – April 2025

- Built an adversarial **agentic AI multi-agent env** where LLMs strategically modified Snake & Ladder boards
- Evaluated **5 SOTA LLMs**; designed metrics for LLM gameplay evaluation, including failure rate, aggression ratio, action distribution, and strategic placement analysis.

Adversarially-robust Model Compression

Aug 2024 – Nov 2024

- Extended ATMC to PointNet for 3D classification with structured pruning, **reducing FLOPs by 75%**.
- Reduced attack success rate from **100% to 43%** under I-FGSM while keeping **>80%** Top-1 accuracy.

TicTacToe Web App | Microsoft Engage 2020 Project

June 2020 - July 2020

- Developed a **JavaScript-based** web application as part of the Microsoft Engage 2020 mentorship program.
- Implemented an **opponent system** allowing users to play against a computer with adaptive difficulty.
- Integrated the **Minimax Algorithm** to optimize the computer’s moves based on selected difficulty levels.